

Timothy Walker

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GitHub: github.com/timwalkercs | LinkedIn: linkedin.com/in/timwalkercs

Portfolio Site: timwalkercs.github.io

EDUCATION

Rutgers University - New Brunswick NJ

2018 - 2022

Bachelor of Science (B.S.) in Computer Science

TECHNICAL SKILLS

Languages: Java, C#, Javascript, Python, HTML, CSS, SQL, T-SQL

IDEs: Visual Studio, Visual Studio Code, Eclipse, Android Studio

Database Tools: MySQL, MS SQL Server, Azure Data Studio

Other: React, .NET Core, ASP.NET Core, Node.js, REST API, Agentic and Generative AI Tools, Azure, GitHub

EXPERIENCE

AI Training Consultant (Software Engineering) | Outlier

January 2024 - April 2026

- Authored reference solutions and step-by-step reasoning for complex programming tasks in **Java**, **Python**, and **C#**, producing training data used to improve frontier language models' code generation.
- Reviewed and ranked model-generated code for correctness, efficiency, and edge-case handling, rewriting failing outputs into working solutions with diagnoses of the underlying error.
- Designed adversarial prompts, targeting known model weaknesses, that fed into subsequent training iterations.

PROJECTS

LINKS

Front-End Application Development | Real-Time Voice & Chat Client

[GitHub](#) | [Website](#)

- Developed and designed the frontend for a Discord-style desktop client supporting live voice, video/screen sharing, and text chat, built with **Electron** and **React** and packaged for distribution with electron-vite.
- Engineered real-time voice and video features over **WebRTC** (mediasoup SFU), including a shared Web Audio mixing graph, RNNoise noise suppression, thorough volume controls, screen sharing with a native source and quality picker, and global OS-wide mute/deafen hotkeys.
- Built a full-featured chat experience with markdown rendering, syntax-highlighted code blocks, emoji, file/image attachments, message editing, typing indicators, and infinite scroll-back history, synchronized via a **WebSocket** event layer with heartbeat and session resumption.

Full-Stack Development | Mechanical Keyboard Switch Database

[GitHub](#) | [Website](#)

- Built a full-stack web application to help keyboard enthusiasts navigate a custom database of mechanical keyboard switches with varying specifications that affect the typing feel and sound profile.
- Designed a responsive **React** frontend with dynamic filtering across 300+ switches, gallery-style browsing, detail views, and state-driven UI updates.
- Engineered a **RESTful ASP.NET Core API** backend powered by **Entity Framework Core** and **SQL Server**, enabling structured data management and flexible query handling for switch specifications.
- Deployed the frontend, API, and database infrastructure through **Microsoft Azure** cloud services.

Embedded AI / Computer Vision | 3D Printer Failure Detection System

[GitHub](#)

- Built a real-time 3D print failure detection system in **Python** that monitors a camera stream over the network and can remotely pause the print when a failure is detected, with a consecutive-frame debounce to prevent single bad frames from halting a valid print.
- Trained and optimized multiple **TensorFlow/Keras** models based on MobileNetV2 to classify normal vs. failed prints under real-world conditions, empirically validated model polarity and confidence thresholds against live prints rather than relying on validation accuracy alone.
- Included a lightweight, local, **web interface** for live camera streaming and frame analysis, outputting "Normal" or "Failure" predictions in real time.